

Supplementary Materials (Unconsciously Forget: Mitigating Memorization Without Knowing What is being Memorized)

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A. Flow Matching

Rectified flow models (Liu, Gong, and Liu 2023; Liu, Gong et al. 2023) aim to train a neural network to define a velocity field through an *ordinary differential equation* (ODE), expressed as:

$$\frac{d}{dt}\psi_t(\mathbf{x}) = u_t(\psi_t(\mathbf{x})), \quad (5)$$

where $\psi_t(\mathbf{x})$ represents the continuous transformation of a data sample $\mathbf{x}$ over time t , describing its trajectory through the learned flow, $u_t(\mathbf{x})$ is the time-dependent velocity field, which dictates the direction and magnitude of the transformation at any point $\mathbf{x}$ in the space. The main objective of the model is to learn $u_t(\mathbf{x})$ that ensures smooth and efficient transport between the *source distribution* $p_0(\mathbf{x})$ and the *target distribution* $p_1(\mathbf{x})$. This transformation follows an ODE over the time interval $t \in [0, 1]$, subject to the following boundary conditions:

$$\psi_1(\mathbf{x}) = \mathbf{x}_1 \sim p_1(\mathbf{x}), \quad \psi_0(\mathbf{x}) = \mathbf{x}_0 \sim p_0(\mathbf{x}). \quad (6)$$

where $\psi_0(\mathbf{x})$ and $\psi_1(\mathbf{x})$ denote the transformed states of $\mathbf{x}$ at times $t = 0$ and $t = 1$, respectively. The optimization objective is defined as :

$$\mathcal{L}(\theta) := \mathbb{E}_{t, \mathbf{x}_t | \mathbf{x}_1, \mathbf{x}_0} \left[\|u_t(\mathbf{x}_t, \mathbf{x}_1) - u_\theta(\mathbf{x}_t, t)\|_2^2 \right], \quad (7)$$

where $u_\theta(\mathbf{x}, t)$ denotes the learned neural network model used to predict the flow at any given timestep t and $u_t(\mathbf{x}, t)$ is the pretrained network.

B. Flow Matching for Image Generation

The image generation process begins by sampling a latent noise vector from a Gaussian distribution, $\mathbf{z}_0 \sim \mathcal{N}(0, \mathbf{I})$. This latent variable $\mathbf{z}_0$ is then evolved through discrete ODE integration. At each time step t_i , the update rule is given by:

$$\mathbf{z}_{i+1} = \mathbf{z}_i + \Delta t \cdot u_\theta(\mathbf{z}_i, t_i, c), \quad (8)$$

where Δt is the time step size, and c represents the text condition. The integration proceeds iteratively, updating the latent variable at each time step until the final latent variable $\mathbf{z}_N$ is obtained at step N through:

$$\mathbf{z}_N = \mathbb{F}_{u_\theta}(\mathbf{z}_0, N, c) = \mathbf{z}_0 + \sum_{i=0}^{N-1} \Delta t \cdot u_\theta(\mathbf{z}_i, t_i, c), \quad (9)$$

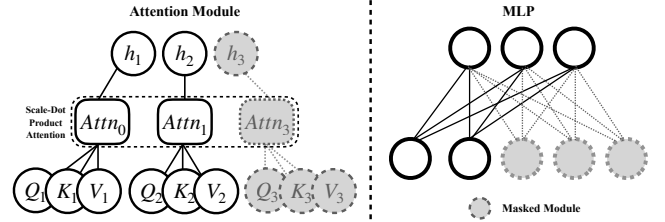


Figure 1: Pruning Masking for attention module and linear layers.

where $\mathbb{F}_{u_\theta}$ denotes the entire sampling process for acquiring the denoised latent $\mathbf{z}_N$. Finally, the trained decoder network $\mathcal{D}$ transforms the final latent variable into an image: $\mathbf{x}_{\text{image}} = \mathcal{D}(\mathbf{z}_N)$.

C. Mask Relaxation

The hard discrete function in Section 3.2 for mask training is defined as follows:

$$\bar{\mathcal{M}} = \sigma \left(\frac{\log(\mathcal{M} + \delta_{\text{re}}) - \log(1 - \mathcal{M} + \delta_{\text{re}}) + \lambda}{\alpha} \right) \quad (10)$$

$$\times (\zeta_{\text{re}} - \gamma_{\text{re}}) + \gamma_{\text{re}},$$

$$\tilde{\mathcal{M}} = \min(1, \max(0, \bar{\mathcal{M}})). \quad (11)$$

where σ denotes as sigmoid function and the parameters are set to $\delta_{\text{re}} = 0.5$, $\zeta_{\text{re}} = 1.1$, and $\gamma_{\text{re}} = -0.1$ during training. To distinguish between the different masks utilized in this work, the mask $\bar{\mathcal{M}}$ in Equation 4 is employed for loss regularization. In contrast, $\tilde{\mathcal{M}}$ serves as the relaxed mask applied to the model module, as illustrated in Figure 1.

D. Training Implementation Details

The default training configuration in this paper is defined as follows:

E. Evaluation Implementation Details

To evaluate the performance of the copyrighted content de-memorization, we utilize SOTA VLM models, GPT-4o (Achiam et al. 2023) and LLaVA 1.6 (Liu et al. 2024).

Table 1: Standard Training Configuration for Unlearning

Hyperparameter	Value
Batch size	20
Learning rate r_{ffn}	0.5
Learning rate r_{norm}	0.5
$\beta_{\text{weak}}, \beta_{\text{medium}}, \beta_{\text{strong}}$	1, 2, 5
Optimizer	Adam
Training steps	2000
Weight decay	1×10^{-2}
Learning rate scheduler	CosineAnnealingLR
Pretrained diffusion model	FLUX.1-Schnell
Hardware setup	$1 \times \text{NVIDIA H100}$

Table 2: Model ID and Version Details of VLMs Used for Evaluation

Model	Model ID	Version
GPT	Azure OpenAI-GPT4o	2024-05-01-preview
LLaVA	LLaVA 1.6	llava-hf/llava-v1.6-vicuna-13b-hf

The detailed version of the GPT-4o¹ and also LLaVA 1.6² are shown as follows:

The evaluation dataset used in this study is sourced from publicly available datasets, including the IMDB 1000 Movies dataset³, the FiveThirtyEight Comic Characters dataset⁴, and the CPDM dataset⁵ (Ma et al. 2024). Due to budget constraints, we randomly selected the most challenging tasks and copyright-sensitive labels, preselected using LLaVA 1.5. We ensured that these sensitive prompts are likely to generate copyrighted content or result in portrait rights violations.

F. Additional Embedding Space Sample of Neutral Prompts

Ablation on Relaxing Sigmoid for Pruning Masking

To evaluate the impact of the relaxed sigmoid function $\hat{\sigma}$, we compare training stability with and without $\hat{\sigma}$, as defined in Equation 4. Figure 3 shows the loss curves under both settings. Without the relaxed sigmoid, training remains feasible and converges eventually. However, the optimization process is volatile, especially in the early stages, often leading to significant oscillations and potential model divergence. The training stabilises by applying $\hat{\sigma}$, reducing variance and ensuring smoother convergence. This stabilization

¹<https://learn.microsoft.com/de-de/azure/ai-services/openai/>

²<https://huggingface.co/llava-hf/llava-v1.6-vicuna-13b-hf>

³<https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows>

⁴<https://www.kaggle.com/datasets/fivethirtyeight/fivethirtyeight-comic-characters-dataset>

⁵<https://github.com/rmpku/CPDM>

Table 3: Compatibility study for target concept removing

Dataset	COCO		Comic		Anime	
β	FID↓	CLIP↑	Acc _g ↓	Acc _t ↓	Acc _g ↓	Acc _t ↓
0.1	32.0	0.30	31.3%	73.8%	78.2%	79.3%
1	32.1	0.29	28.9%	70.2%	36.5%	52.2%
2	32.4	0.28	10.2%	41.5%	30.3%	42.8%
5	38.9	0.25	9.1%	33.3%	16.3%	27.4%
10	47.2	0.23	8.3%	31.3%	15.7%	26.3%
FLUX schnell	32.0	0.30	42.6%	90%	68.7%	77.6%

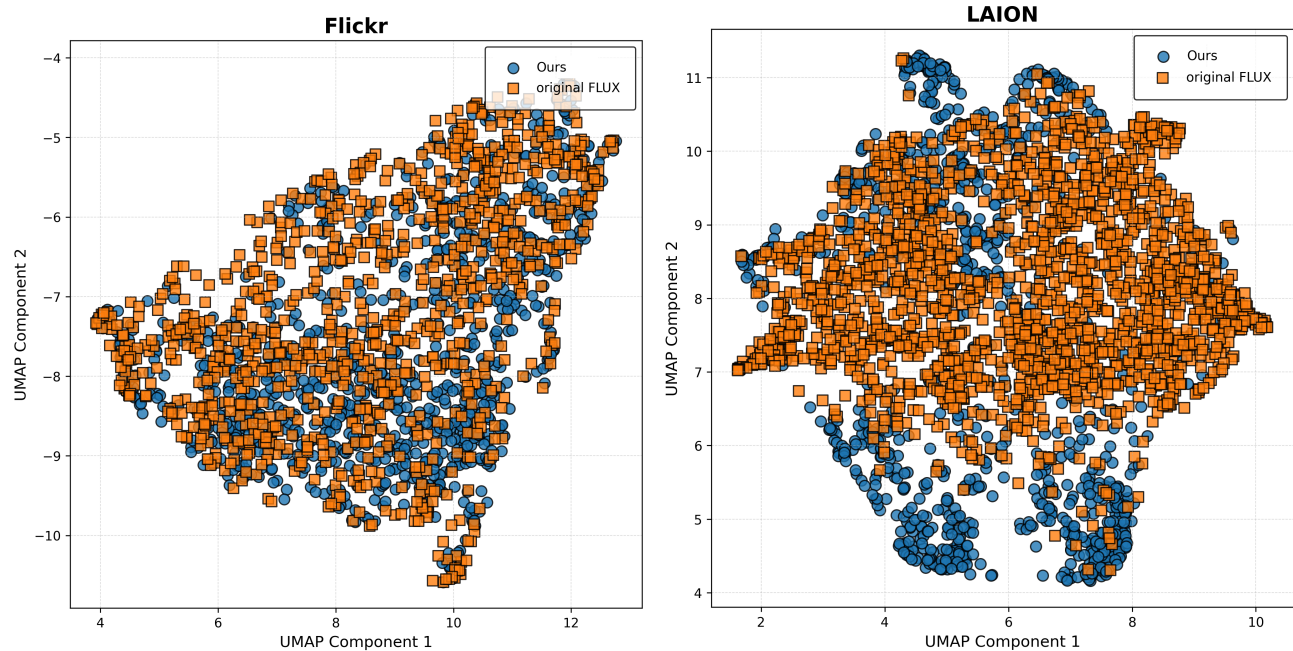
enhances training robustness and leads to more consistent and reliable final model performance.

G. Hyperparameters Ablation Study

Table 3 presents the results of the ablation study on the unlearning coefficient β . We selected 1, 2, and 5 to represent weak, medium, and strong settings, respectively. A lower value of $\beta = 0.1$, preserves high general generative capabilities but is insufficient for effectively de-memorizing copyrighted content. Conversely, when β is set to 10, the general generative performance deteriorates significantly.

References

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(a) Neutral Prompts: Flickr

(b) Neutral Prompts: LAION

Figure 2: Additional visualizations of the embedding space using UMAP.

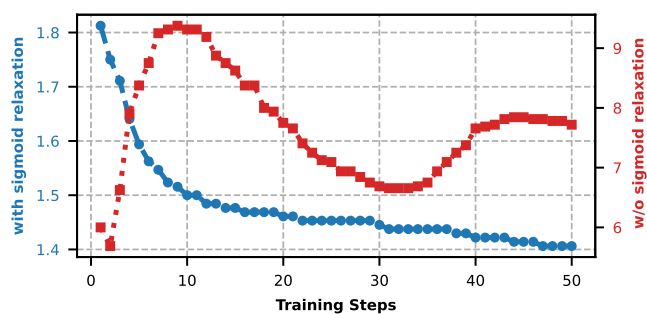


Figure 3: Regularization Loss w/ or w/o Sigmoid Relaxation. This plot shows the impact of sigmoid relaxation on the loss term to enable smooth training.